ECE 424 — Engineering Professional Practice

Spring 2026 Syllabus

Course Information

Instructor: Branimir Pejcinovic, pejcinb@pdx.edu

Office Hours: Thursdays, 15:15 – 16:15, office or zoom

Times and location: Fridays, 15:30 – 17:10, EB 92

Platform: Canvas LMS

Course Description

Bulletin description: Prepares graduates for careers in electrical and computer engineering. Topics include ethical reasoning and considerations, strategies for job acquisition, career planning, certification and licensure, approaches to lifelong learning, and means of maintaining awareness of contemporary global and local societal issues.

Course Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Communicate effectively in professional engineering contexts, including written documents, emails, and oral presentations, to audiences of varying technical levels.
2. Develop and articulate a career plan using structured self-assessment frameworks informed by current labor market realities.
3. Describe the fundamentals of intellectual property, including the patent process and its relevance to engineering practice.
4. Recognize ethical problems in engineering practice and apply structured analysis methods to evaluate and respond to them.
5. Describe the role of ethics in design decisions
6. Analyze the ethical implications of AI and autonomous systems in engineering and professional responsibility in the age of generative AI.
7. Acquire new knowledge and recognize the importance of lifelong learning and professional development as an ongoing responsibility of engineering practice and career development.
8. Describe current social, environmental, political and economic factors influencing the development and use of technology

ABET Student Outcomes Addressed

Outcome 3: An ability to communicate effectively with a range of audiences.

Outcome 4: An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

Outcome 7: An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Required Materials

Textbooks and Readings

- **A. Fasano,** "Engineer Your Own Success" — available free through PSU library. Chapters 1, 3, and worksheets 11-3, 11-4, 11-5. Available as e-book: https://onlinelibrary-wiley-com.proxy.lib.pdx.edu/doi/book/10.1002/9781118886519
- **S. Ramanathan,** "Critical Thinking Skills for Engineers — Book 2: Communication Skills" — Canvas.
- **Selinger,** "Stuff You Don't Learn in Engineering School" — ch. 12 on ethics, Canvas.
- **Bazerman & Tenbrunsel,** "Blind Spots: Why We Fail to Do What's Right and What to Do about It" — Chapters 1–3, available as e-book through PSU library: https://ebookcentral.proquest.com/lib/psu/detail.action?pq-origsite=primo&docID=664630
- **IEEE Code of Ethics** (2020 version) — https://www.ieee.org/content/dam/ieee-org/ieee/web/org/about/corporate/ieee-code-of-ethics.pdf and https://www.ieee.org/about/corporate/governance/p7-8 .
- Additional readings, articles, and videos will be posted on Canvas throughout the term.

Assessment and Grading

Component	Weight
Homework (6 assignments, use 5 best)	25%
Canvas Discussions (2 graded)	10%
Articles & Presentations (7 weekly)	10%
News Presentation (team)	15%
Career Plan Project	20%
Final Exam (comprehensive)	20%
Total	**100%**

Note: Point allocations are approximate and may be adjusted. Any changes will be announced on Canvas.

Grading Scale

A: 90–100% | B: 80–89% | C: 70–79% | D: 60–69% | F: below 60%

Plus/minus grades may be assigned at the instructor's discretion.

Weekly Schedule (draft – subject to change)

Week	Topic	Key Activities	Due This Week
1	Audience Analysis & Written Communication	Intro lecture; Audience framework; email writing; AI & writing; assign Welcome to Earth activity	A&P1 (intro)
2	Presentations & Spontaneous Speaking	Welcome to Earth presentations; slide craft; poor presentation examples; spontaneous speaking & anxiety; impromptu exercise	A&P2 (presentations); Discussion1 (Comms); HW1 (survey)
3	Career Planning: Job Market & Self-Assessment	Tektronix video discussion; AI-era job market lecture; personal SWOT analysis in-class activity	Canvas quiz (Tek); A&P3 (jobs) Worksheet 11-3; HW 2 (email)
4	Career Development: SMART Goals & Professional Growth	SMART goals lecture & in-class activity; career trajectories; networking; career plan workshop	A&P4 (SWOT); Discussion2 (Presence) Worksheet 11-4 HW 3 (job search)
5	Ethics Part 1: Foundations	Therac-25 case; trolley problem & frameworks; IEEE Code of Ethics; 7-step method; in-class application	A&P5 (TBD) HW 4 (metacognitive)
6	Ethics Part 2: Behavioral & Social Ethics	Bazerman & Tenbrunsel Blind Spots framework; Ariely scandal case study; team scenario exercise (50 min)	A&P6 (TBD) Discussion3 (ethics); A&P #6;
7	Ethics Part 3: AI Ethics	O'Neil TED talk; A/IS to LLMs history; hallucination, deepfakes, AI code, bias; IEEE 7000; professional responsibility	A&P7 (TBD) HW 5 (ethics)
8	News Presentations	All team news presentations; peer feedback and debrief	News presentation slides & delivery
9	IP & Patents (Guest Lecture)	What and why of IP and patents	Career Plan Project due;

10	Lifelong Learning, Sustainability & Course Wrap-Up	Lifelong learning; sustainability in engineering; course synthesis and exam review	HW6 (patents) Exam study guide distributed
Finals	Final Exam (Comprehensive)	Covers all 10 weeks of content; in person	3:40-5:30pm, Tuesday, June 9

Note: Schedule is subject to adjustment. Any changes will be announced on Canvas with advance notice.

Teams

Students will be assigned to teams of 3–4 (approximately 10 teams). Teams remain consistent throughout the term and are used for news presentations, in-class scenario exercises, and discussion subtopic assignments. Team formation occurs in weeks 1 and 2.

Policy on AI Tools

AI writing and coding tools (ChatGPT, Claude, Copilot, Grammarly, etc.) are a reality of professional engineering practice, and this course acknowledges that. You are permitted to use AI tools to assist with drafting, structuring, grammar-checking, and brainstorming for assignments. However:

- The reasoning, ethical judgment, and final content must be your own.
- If you use AI tools, **briefly disclose** what you used and how (e.g., “I used Claude to help structure my outline”). This is a transparency requirement, not a penalty.
- Submitting AI-generated content as your own original analysis without disclosure is a violation of academic integrity and, fittingly, an ethics violation in a course about professional ethics.
- You own what you submit. AI tools do not transfer responsibility for accuracy or quality.

Attendance and Participation

This is a fully in-person course. Attendance is expected at all sessions and contributes to your participation grade. Several weeks include in-class activities that cannot be replicated asynchronously. If you must miss a class, notify the instructor in advance.

Academic Integrity

All students are expected to adhere to PSU’s Student Code of Conduct and Academic Integrity policies. Plagiarism, fabrication, and cheating are serious violations and will be handled in

accordance with university procedures. In a course about professional ethics, academic integrity is not merely a policy — it is a core learning outcome.

Accessibility and Inclusion

Portland State University is committed to providing equal access to all students. If you have a disability or condition that may require accommodations, please contact the Disability Resource Center (DRC) to register and coordinate services. The instructor is committed to creating an inclusive learning environment and welcomes communication about how to best support your learning.

This course values diverse perspectives and experiences. The ethics discussions, in particular, benefit from the range of cultural, professional, and personal viewpoints that students bring. All students are expected to engage respectfully with peers who hold different views.

Various PSU policy statements and resources

Academic Integrity & Grading Policies	**Student Support Resources**
• PSU Academic Calendar • PSU Grading System • Student Code of Conduct • Incomplete Grades Policy	• How to Find Help at PSU • Access and Inclusion for Students with Disabilities • Understanding Sexual Misconduct • Title IX Reporting • Religious Accommodations

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In Case of Emergency	**Call *503*-725-4407 for Non-Emergencies**
Call 9-1-1 for Emergencies **Immediate threat to life and safety** *For issues such as: a medical emergency, urgent violent incident, fire, etc.* **You can also call 503-725-5911**	Campus Public Safety Office (CPSO) **– Non-Emergency** *For non-emergency issues such as: vandalism, disturbance, suspicious person, theft, suspicious packages, access control, etc.*